Lab 7: Fuzzing for Input Validation Bugs

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Assigned repo: SETAPESU26/L6_L7_32_weather_alert (lab7/)

Submission repo: github.com/SUKRUTH-ES/UE24CS341A-Software-Engineering (Lab7/)

Script under test: weather_alert.py (the handout's processor.py) with parse_reading, wind_category and rainfall_spread. Tools: Python 3.14.0, pytest 9.2.0, Hypothesis 6.170.15.

Summary

	Original script	Fixed script
Fuzz tests (test_weather_alert.py)	7 failed, 3 passed	10 passed
Distinct bugs found by Hypothesis	14 (10 crashes, 4 wrong results)	0
python weather_alert.py with input 'hello'	crashes: ValueError traceback	prints 'Invalid reading: ...' and continues

Bugs found by fuzzing (minimal failing inputs)

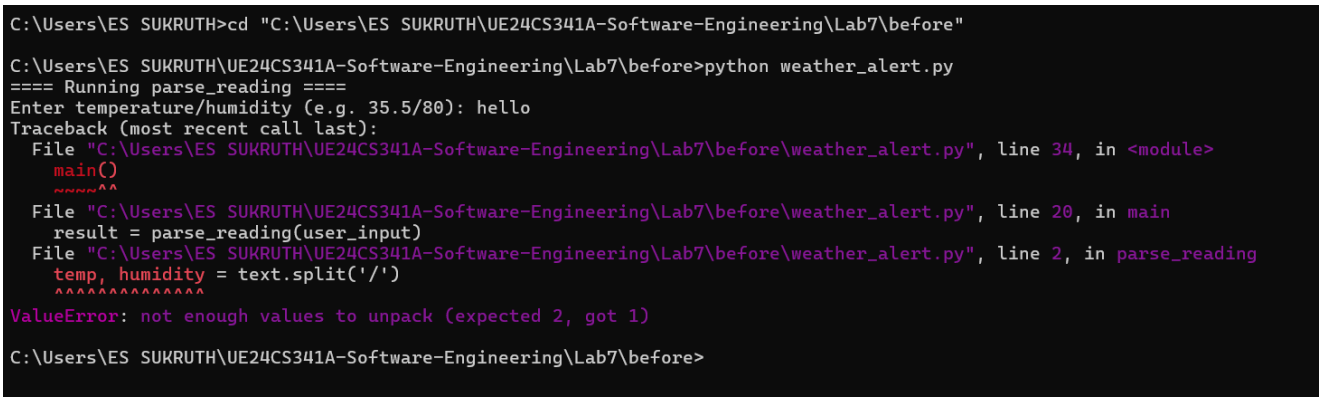
Function	Input	Original behaviour
parse_reading	None / '' / '/' / b''	AttributeError / ValueError (unpack) / ValueError (float) / TypeError
parse_reading	'0/-1', 'NAN/0'	accepted humidity -1 and temperature nan (wrong result)
wind_category	None / ''	TypeError / ValueError
wind_category	'-1'	graded a negative speed as 'none' (wrong result)
rainfall_spread	None / ''	AttributeError / ValueError
rainfall_spread	'NAN', 'INF'	returned nan as the spread (wrong result)
all three	non-strings (int, list, bytes...)	AttributeError / TypeError
main()	any of the above typed at a prompt	whole program stops with a traceback

Fix

Defensive validation in every function: check the type, split into the expected number of parts, convert inside try/except (ValueError/TypeError only), reject nan and inf, and check ranges (humidity 0-100, temperature -100 to 100 C, wind 0-500 km/h, rainfall >= 0). Invalid input returns None instead of raising, and main() prints a clear message. Valid input gives exactly the same output as before.

Step 2: Run the original (buggy) application

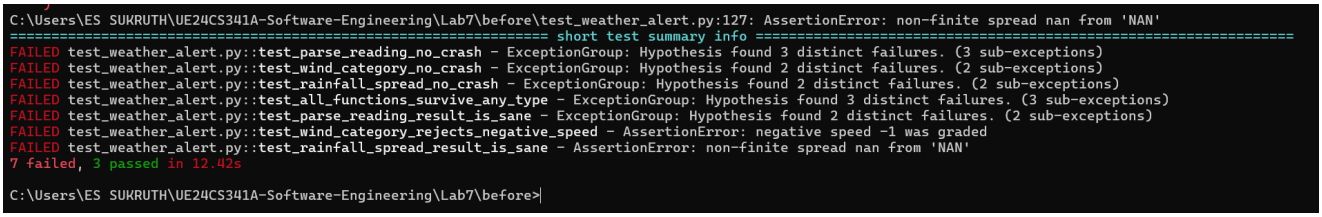
```
C:\...\Lab7\before> python weather_alert.py   (input: hello)
```



The original parse_reading assumes the input contains exactly one '/', so a reading without one crashes the whole program with a ValueError.

Step 4: Run the Hypothesis tests against the original code (failing)

```
C:\...\Lab7\before> python -m pytest test_weather_alert.py -q --tb=line
```



7 of 10 tests fail. Hypothesis reports several distinct failures per test and shrinks each to a minimal input, for example '' and None (crashes), -1 (negative wind speed graded), and 'NAN' (nan accepted as a rainfall spread).

Step 5: Run the fixed application

```
C:\...\Lab7> python weather_alert.py   (inputs: hello, -5, 1,,3)
```

```
C:\Users\ES SUKRUTH\UE24CS341A-Software-Engineering\Lab7>python weather_alert.py
==== Running parse_reading ====
Enter temperature/humidity (e.g. 35.5/80): hello
Invalid reading: use temperature/humidity, e.g. 35.5/80 (humidity 0-100)

==== Running wind_category ====
Enter a wind speed in km/h (e.g. 45): -5
Invalid wind speed: enter a whole number from 0 to 500

==== Running rainfall_spread ====
Enter hourly rainfall in mm as CSV (e.g. 2,5.5,10): 1,,3
Invalid rainfall list: enter numbers >= 0 separated by commas, e.g. 2,5.5,10

C:\Users\ES SUKRUTH\UE24CS341A-Software-Engineering\Lab7>
```

The same kind of invalid input is now rejected with a clear message at each prompt, and the program finishes normally.

Step 6: Re-run the Hypothesis tests against the fixed code (passing)

```
C:\...\Lab7> python -m pytest test_weather_alert.py
```

```
C:\Users\ES SUKRUTH\UE24CS341A-Software-Engineering\Lab7>python -m pytest test_weather_alert.py
============================= test session starts ==============================
platform win32 -- Python 3.14.0, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\ES SUKRUTH\UE24CS341A-Software-Engineering\Lab7
plugins: hypothesis-6.170.15, platformdirs-4.14.0
collected 10 items

test_weather_alert.py ..........                                         [100%]

============================== 10 passed in 14.62s ==============================

C:\Users\ES SUKRUTH\UE24CS341A-Software-Engineering\Lab7>
```

All 10 tests pass: the no-crash fuzz tests (up to 500 generated inputs each) and the behaviour properties (correct results for valid input, no nan/inf, values in range).